



How feathers evolved

Researchers are studying chickens, ostriches, and penguins to learn how flight feathers evolved. <https://digit.in/dec19-59>



CO2 consuming bacteria

Laboratory-evolved E.Coli strains have learned to consume carbon dioxide for growth. <https://digit.in/dec19-60>

SWEAT SENSORS

Bid farewell to needles

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ven for adults, one of the scariest aspects of a visit to the doctor, is the use of needles to

withdraw blood samples for medical tests. Once the sample is collected, it is sent to a testing facility, that analyses the sample, and then sends back a report. By this time, there might be changes in the body of the

patient. Wouldn't it be great if there was a non invasive way to continuously collect real-time information on the physiological condition of a patient? Turns out, the answer may lie in wearable sweat sensors.

Sweat can be more easily sampled than say tears, saliva or intestinal fluids. By measuring the concentration of biomarkers such as glucose, cortisol and sodium present in the sweat, sensors can derive information on diseases, dehydration, fatigue

or stress in an individual. In fact, there are actually dedicated sweat test centers, but in this case, collecting the sweat and sending the samples for testing is an even more laborious process than a blood test. The patient has to sit still for as long as it takes for the sampling vial to fill up with sweat, which can take up to half an hour. The process is particularly wearisome for children.

The challenge is to miniaturise the testing apparatus. Some of the approaches being used include wristbands, armbands and stickers. The advancements needed to make viable sweat sensors that are sensitive enough while working with tiny power supplies, flexible electronics, and advancements in microfluidics, which involves precisely handling small amounts of liquids using the tinnest of pipes. All of these are coming together in research institutions around the world, and viable sweat sensors are being realised. The sweat sensors can capture the sweat using absorbent pads. A variety of materials can be used to embed the flexible electronics, including plastics, fabrics and elastomers.

THE TECHNOLOGY

Between 2016 and 2017, a collaborative team of researchers from UC Berkeley and Stanford combined two existing technologies to create a non invasive sweat sensor. These were a flexible circuit board, and a flexible array of sensors. The two were put together in a neat little wristband, and could analyse the physiological condition of the wearer at a molecular level. Till that point of time, non invasive sensors could only measure the concentration of a single molecule at a time. By using an array of sensors, the team developed what they call a perspiration analysis system, that could be placed onto the wrist of any patient. The band was designed to stick to the skin. The circuit board provided the signal processing, and wirelessly transmitted the collected data to a paired smartphone.